

# **Volatility Dynamics in Indian Futures Trading**

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# **Volatility dynamics in Indian futures trading**

## **Abstract**

We estimate realized volatility using intraday interval data on gold and crude oil futures trading at the MCX, the leading commodities exchange in India for approximately three years, January 2006 to December 2008. We consider two measures of daily volatility, sum of squared returns and Parkinson (1980) range based estimator for our volatility estimates. We test for and identify short term periodicity and long term seasonality effects in realized volatility. We further determine the relation between volatility and time to expiration (Samuelson's maturity hypothesis) for those futures contracts after controlling for appropriate seasonality effects. Our results show strong evidence of intraday periodicity, bimonthly/quarter-of-the-year seasonality, and a net inverse relation between time to maturity and realized volatility even after controlling for spot volatility and net order flow denoting information and inventory uncertainty.

JEL Classification: G13

*Keywords: gold and crude oil futures, intraday, volatility, maturity hypothesis*

## **I. Introduction**

We estimate daily volatility from intraday trade data on gold and crude oil futures trading at the Multi Commodity Exchange (MCX), the leading commodities exchange in India,

and identify periodicity and seasonality in the estimated realized volatility for both futures. Further, we test Samuelson's maturity hypothesis that predicts an inverse relation between volatility and the time to expiration for those two futures contracts after controlling for appropriate seasonality effects, spot volatility, and net order flow denoting multiple volume related effects including information and inventory effects on volatility. Our sample period spans from January 2006 to December 2008.

Organized futures trading have existed in India for a long time but the current structure of the futures market in India is due to the liberalization and transformation of the financial markets that began in 2000.<sup>1</sup> Currently there are 23 organized futures exchanges in India out of which three, National Multi-commodity Exchange (NMCE), Multi Commodity Exchange (MCX), and National Commodity and Derivatives Exchange (NCDEX) are national exchanges. The Forward Market Commission (FMC), the regulatory body of the commodity futures markets in India lists, as of March 2007, 153 different commodities contracts traded on the various commodity exchanges in India; however, only a few of those products are actively traded.

Regarding aggregate market liquidity, as of March 2007, the value of traded futures in India is US \$3.4 billion, 27 (73) percent of which are agricultural (non-agricultural) commodities. Table 1 below lists the top commodity futures as of 2009 in terms of trading volume. Table 1 indicates that on the basis of trading volume, metals and energy are a few of the most actively traded commodity futures in the Indian market and MCX is the market leader in those commodity futures. Table 1 further points at the

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<sup>1</sup> Shah and Gorham (2008) contains an excellent history and structure of the market and reports that the first commodity futures traded on Indian soil is a cotton contract in 1875.

relative market share of each future with respect to the exchange portfolio based on the value of traded contracts. Gold and oil futures are two most liquid contracts and account for more than half of MCX's trading portfolio.

In this paper, using intraday interval data obtained from MCX, we specifically examine the following two empirical questions:

- 1) Is there seasonality e.g., day of the week and/or month of the year and/or periodicity i.e., time of the day effects in realized volatility?
- 2) Does volatility increase around contract expiration dates- Samuelson's maturity effect hypothesis? Further, does the relation between volatility and time to expiration exist after controlling for spot volatility and net order flow as proxies for information and inventory uncertainty?

Our hypotheses regarding volatility are grounded in information economics and market microstructure theories. Volatility denotes information flow that is considered random and unpredictable. Recurring and consistent intraday patterns, periodicity in volatility may indicate certain timeliness of information flow, which may then be used to argue against a random information process. Market microstructure theories argue that the time to trade is a strategic option for traders and may reveal information about the future value of a security (Diamond and Verrecchia [1987], Easley and O'Hara [1992]). If these were to be true, profitable trading strategies based on intraday timing of trades may be constructed and exploited. Further, seasonality in volatility evidenced by day of the week and/or month of the year effects in time series behavior indicate predictability and thus refutes efficient market pricing of volatility risk. Such seasonality effects must

be controlled for or removed before any econometric model is tested, otherwise the results might be spurious.

Second, Samuelson's hypothesis on the inverse relation between futures price volatility and the contract expiration dates, also known as the maturity effect has been widely tested with respect to a range of futures products in many diverse markets over a long period of time.<sup>2</sup> Empirical evidence for example, in Milonas (1986) and Bessembinder et al. (1996) points out that while the relation generally holds for agricultural and other commodity futures, the relation falls short in the case of financial futures, where there is no physical inventory holding cost. Hence, Bessembinder et al. (1996) posit that the physical carrying cost of commodities is the underlying justification for Samuelson's hypothesis to hold for commodities but not financial futures markets and formulates a negative covariance hypothesis that tests an inverse relation between changes in periodic carrying cost and spot price. Nevertheless, empirical evidence on Samuelson's hypothesis and the complementary negative covariance hypothesis on the basis of intraday data is rare and thus we have a fairly unique opportunity to test those hypotheses on the basis of intraday data for an emerging Indian futures market. In addition, testing maturity hypothesis for gold and oil futures in particular offer interesting possibilities since those futures are not perishable like agricultural futures and as such do not have similar carrying cost but may contain substantial inventory and asymmetric information risk. In this paper, we examine those issues with respect to an extended maturity risk hypothesis that tests both a linear and a non-linear relation between volatility and time to expiration in the presence of inventory and information asymmetry.

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<sup>2</sup> See Duong and Kalev (2008a, 2008b) for a brief summary.

Our empirical results are as follows. First, the empirical distributions of two volatility estimates, sum of squared returns and Parkinson's range measure indicate that the latter is downward biased and less efficient for both gold and crude oil futures. Second, both gold and crude oil futures volatility estimates show evidence of almost similar kind of periodicity but different forms of seasonality. Finally, we find evidence of a net inverse non-linear relation between realized volatility and time to maturity for gold and crude oil futures. However information and inventory uncertainty denoted by spot volatility and absolute value of net order flow explain a much greater part of the variation in realized volatility.

Our research contributes to the existing literature in several important ways. The Indian futures market in its current form is a rather new and evolving marketplace and perhaps due to the infancy of its present form, serious academic studies on the Indian futures market are limited.<sup>3</sup> The absence of reliable data on the subject has further compounded the problem. We obtain exclusive proprietary data from MCX, the leading commodities futures exchange in the nation on two of its most liquid futures, gold and crude oil futures and hence we limit our investigation to those two futures only. In addition, unlike in equity and options markets, intraday studies related to futures markets are rare even for developed and mature markets and from that perspective, this is the first and only research study so far on Indian futures market using intraday data on two

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<sup>3</sup> Only a handful of published studies on Indian futures market exist which are in essence market efficiency studies and have looked at issues of cross market information flow between stock and futures, and spot and futures markets specifically for index futures. Vipul and Jacoby (2007) and Jacoby et al. (2008) investigate the relative efficiency of range based estimates of volatility for Nifty 100.

individual commodity futures.<sup>4</sup> Finally, we extend the empirical model to test Samuelson's maturity hypothesis by introducing a non-linear trend and net order flow denoting information and inventory effect on volatility, which are critically important for gold and oil futures. We find empirical results supporting our extended model.

The paper proceeds as follows. In section II, we provide a brief literature review on volatility in futures markets. We describe the data, discuss volatility estimation, and test intraday periodicity and other seasonality effects in volatility in section III. Section IV provides a framework to test Samuelson's hypothesis on the inverse relation between volatility and time to maturity and discusses the empirical results. Section V concludes.

## **II. Literature review**

This study builds on two separate lines of research in finance- one, volatility estimation from intraday data and two, Samuelson's maturity hypotheses testing in the context of information and inventory risk in an emerging futures market. In this section, we provide a brief literature review of these two main strands of literature with specific references, whenever possible, to the empirical evidence related to gold and crude oil futures.

While volatility is an important measure of individual securities and financial market performance, it is not observable and thus statistical and econometric models are proposed to estimate volatility with minimum measurement and/or forecast error. With intraday rather than daily and quotes instead of transaction price data, the estimation

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<sup>4</sup> Intraday studies on futures markets include Duong and Kalev (2008), Mann (JB), Martens (2005), and Daigler (1997). Fung and Tse (2008) conduct an intraday analysis of SSF (single stock futures), an innovative product and the underlying stocks traded in the Stock Exchange of Hong Kong.

process has become both rich and complex. Anderson and Bollerslev (1998) find from a study on multiple currency products that volatility estimated as sum of squared returns computed from two successive bid/ask quotes at five minutes intervals are unbiased and have the lowest mean squared error. Daigler (1997) uses standard deviation and Glassman-Klass (1980) range measure as estimators of realized volatility and reports strong evidence of intraday periodicity in both of those estimates of volatility. Martens (2005) confirms the lowest forecast error when volatility is estimated by sum of squared returns over five minutes intervals over several other interval estimates for S&P index futures. Bianco and Reno (2006) observe periodicity in serial autocorrelation in intraday returns for Italian index futures, FIB30.

In terms of Samuelson's maturity hypothesis, empirical evidence based on low frequency daily data from multiple future markets conclusively finds that agricultural and perishable commodity futures tend to exhibit an inverse relation with time to maturity while financials generally do not (see for example, Grammatikos and Saunders, 1986; Khoury and Yourougou, 1993; Galloway and Kolb, 1996; Bessembinder et al, 1996).<sup>5</sup> Bessembinder et al (1996) find support for their hypothesis that agricultural and commodity futures have high negative covariance between spot price changes and carrying cost, which they predict as the underlying cause of the evidence in support of maturity hypothesis for agricultural futures. Recently, Duong and Kalev (2008a, 2008b) provide evidence on Samuelson's hypothesis and negative covariance hypothesis using

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<sup>5</sup> Chen et al (1999) report that for Nikkei 225 index futures volatility decreases around the time of expiration and Barnhill (1987) finds maturity effect in US t-bond futures.

multiple measures of volatility computed from intraday data on several commodities and financial futures.

Anderson and Danthine (1983) propose a state variable hypothesis in which any proxy for information risk e.g., spot volatility is the primary determinant of volatility. Anderson (1985) and Kenyon et al. (1987) find seasonality to be more informative than maturity for the determination of price volatility for several agricultural futures. This state variable hypothesis might be particularly relevant for gold and crude oil futures. Agricultural commodities are perishable and thus those commodity futures bear a substantial carrying cost which is not true for financial futures. Metals including precious metals and energy are not perishable and thus do not have similar risk as agricultural commodities. However, because of the inherent hedging and speculative attributes of the underlying commodities, gold and oil futures contain substantial inventory and price risk and thus we would expect spot volatility and net order flow or bid ask spread to have a more significant impact on volatility than maturity and seasonality. Understandably, there are differences between gold and oil futures in terms of those attributes, which will be evident from the empirical results.

Indeed, prior empirical evidence on gold and crude oil futures bears out our contention that those futures do not share many common risk attributes with agricultural and financial futures. From an intraday analysis, Duong and Kaley (2008a) find only mixed evidence regarding oil and gold futures. They find support for maturity hypothesis in the case of TOCOM crude oil but TOCOM gold and NYMEX oil futures do not support the maturity hypothesis. Further their GARCH model testing for all the three oil

and gold futures fails to support maturity hypothesis. On the contrary, their testing of maturity hypothesis after controlling for spot volatility find spot volatility a significant determinant of realized volatility for all three gold and oil futures while TTM is significant only for gold futures. With realized range as a measure of volatility, Duong and Kalev (2008b) find evidence in support of maturity hypothesis only for agricultural futures.

### **III. Volatility: Data, estimation, and intraday effects**

#### **III.A. Data and sample statistics**

We obtain market quotes, price, and volume data at 10 minutes interval on gold and crude oil futures, two most liquid futures contracts from January, 2006 to December, 2008 from MCX, the leading commodity futures exchange in India. MCX markets more than 40 commodity futures; however, a majority of those futures are not sufficiently liquid. In this paper, we focus on the two most liquid futures in MCX's portfolio, gold and crude oil.

MCX started its operation in 2003 as an online settlement and clearing entity for all futures transactions in India and in 2005 launched the first commodities futures index, MCX COMDEX. Within six years since its inception, MCX became the fourth largest derivatives exchange in Asia with a record turnover of more than US\$ 6.39 billion. Currently it is one of the largest (6<sup>th</sup> largest commodities exchange in the world in terms of number of traded contracts between January and June 2009) global derivative markets

with a wide range of commodity and index futures products and a national and international network of traders, who trade through its electronic trading system.

Regarding trading, a few of the unique institutional features of the Indian futures market in general and MCX in particular are a) new age fully electronic trading platform for futures trading in India. MCX operates a 14-hours trading day 5 days a week, and an additional 7-hours of trading on Saturdays allowing ample trading opportunities from remote sites, b) absence of market makers and a strict electronic limit book operation, and c) concentration of trading close to expiration and a precipitous drop in trading volume initially when a new contract rolls in.<sup>6</sup> Crude oil contracts are live for a month and hence their current month contracts enjoy sustained liquidity for 30 days prior to the expiration, while gold contracts are live for 60 days and enjoy limited but sustained liquidity for the entire 60 days period till expiration.

Our data set contains the best bid and ask quotes, best buy and sell trade size (number of contracts corresponding to highest bid and lowest ask quotes), and the last traded price for all gold and crude oil futures at every 10 minutes interval for the near month (i.e. one month before expiration for crude oil and two months before expiration for gold futures) and 30 minutes interval for next month (one month for crude oil and two months for gold futures) contracts. At any time there are six crude oil and four gold contracts available for trading in the market. However, actual trading occurs only in the near month and in a substantially reduced way in the next month contracts. There is no trading beyond the next month e.g., a July 15 crude oil contract will have most liquidity during the one month period beginning June 16 (near month) and some volume albeit

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<sup>6</sup> During summer trading hours are a few hours more per week than the rest of the year.

substantially reduced during the period May 16 to June 15; similarly, a July 15 gold contract will see sustained trading volume for the two months period beginning May 16 and substantially reduced volume during the one month period, April 16 to May 15.

MCX operates long trading hours (14-hours trading day 5 days a week, and a 7-hours day on Saturdays) and hence our sample consists of 84 10-minutes intervals 5 days a week and 42 similar intervals on Saturdays for the near month contracts. Due to a precipitous drop in trading volume from the near month to that in the next month, next months' contracts data are available only at 30 minutes intervals. To avoid problems with non-synchronous data, we decide to leave out the next month data and use data only from the near month contracts for our analysis.

Our interest is in volatility which is not observable and thus we need to estimate it. We use two estimators of volatility, sum of squared returns via Anderson and Bollerslev (1998) and a range based estimator proposed by Parkinson (1980). We discuss the volatility estimates and the corresponding descriptive statistics below.

### **Sum of squared returns**

Anderson and Bollerslev (1998) compute daily volatility as sum of squared returns from five minutes interval data and report that it has lowest mean forecast error in multiple foreign exchange markets. However, they also document it to be a noisy estimator of daily realized volatility. Further, they suggest using the bid ask quote midpoint instead of transaction price to account for the bid ask bounce as pointed out by Roll (1984). Other

studies using similar intraday data have used one, five, and ten- minutes interval data to estimate daily realized volatility.

We obtain data at 10 minutes interval and using the data, we compute daily volatility from intraday interval bid and ask prices following Anderson and Bollerslev (1998) and Duong and Kalev (2008) as follows:

Daily volatility,  $\sigma_t^2 = \sum_{i=1}^{81/84} r_t^2$

10 min interval return,  $r_t = \ln\left(\frac{P_t^S + P_t^B}{P_{t-1}^S + P_{t-1}^B}\right)$

We annualize volatility by multiplying daily volatility with 250.<sup>7</sup> Tables 2A and 2B (left panels) contain descriptive statistics on log (annualized daily volatility) computed as above from intraday squared returns for gold and crude oil futures respectively. For robustness check, we compare the above descriptive statistics on daily volatility estimates with those where returns are computed as  $r_t = \ln(p_t/p_{t-1})$ . The descriptive statistics turn out to be mostly identical implying no impact of bid ask bounce and thus we do not report those descriptive statistics.

We find from Table 2A that for gold futures, the mean and median of the annualized volatility estimates are -3.99 and -3.91 respectively indicating those are small numbers close to zero in pre-log values; nevertheless there is sufficient standard deviation of approximately 1.09 for each of the realized volatility estimates to make amenable to

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<sup>7</sup> It is not clear which way is our estimated annualized realized volatility biased- there is an upward bias since Saturdays are excluded but that also reduces the number of days multiple (250) for computing annualized volatility.

modeling. The median volatility estimate is close to the mean implying little asymmetry as confirmed by a low value of skewness (-0.40).

As reported in Table 2B, for crude oil futures, the mean (standard deviation) of the annualized volatility estimates are -2.997 (0.983) indicating those are almost zeros (close to 1) in pre-log values with moderate standard deviations. We do not find any evidence of bid ask bounce in the data and for the remaining analysis we continue only with volatility estimated as sum of squared returns where returns are computed from mid quotes at two successive intervals.

In addition, we provide descriptive statistics on Saturday volatility to highlight the differences between the distributional properties of Saturday and other weekday volatility. Eyeball statistics confirm that the distributions of Saturday volatility and weekday are different. The means and medians for Saturday are approximately two times more negative on a log scale than those of weekdays indicating volatility is much less on Saturdays for both gold and oil futures. The quartiles of the distribution of Saturday volatility are also quite distant from those of the weekday volatility, which confirms the differences in the distributions of volatility for weekdays vs. Saturdays. Nevertheless, we conduct a Kolmogorov-Smirnov goodness of fit test ( $Ksa_{gold} = 0.8739$  p-value  $< 0.000$  and  $Ksa_{oil} = 0.965$  p-value  $< 0.000$ ) and hence conclude that the distributions of weekday and Saturday volatilities are not similar and hence we exclude Saturday observations from further analysis.

### **Range based measure**

Parkinson (1980) assumes a diffusion process without a drift for security prices and from that estimates volatility from daily high and low prices as follows:

$$\hat{\sigma}_p^2 = \frac{1}{4 \ln 2} (\ln H_t - \ln L_t)^2.$$

Glassman and Klass (1980), Rogers and Satchell (1992), and Yang and Zhang (2000) extend Parkinson (1980) by including functions of opening and closing prices of the day and propose several estimators for daily volatility.<sup>8</sup> Almost all of the above estimators are downward biased; however, Parkinson's is also the least efficient (Chou et al., 2005). Nevertheless because of the difficulty of ascertaining opening and closing prices from interval data, we use the Parkinson estimator for this study. We create an intraday time series of mid quotes and determine the highest ( $H_t$ ) and lowest ( $L_t$ ) points in the time series.

Tables 2A and 2B (right panels) contain summary statistics on the Parkinson estimates for gold and oil futures respectively. The distributional properties of the Parkinson estimates for both gold and crude oil futures confirm that compared to sum of squared returns Parkinson's estimates are downward biased (for means and for each quartile) and less efficient (standard deviations are higher).

### **III.B. Testing intraday periodicity and seasonality effects in volatility**

The literature on intraday futures trading is very limited and there are only a few observed patterns or shapes for an intraday realized volatility function in the futures

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<sup>8</sup> Please refer to Shu and Zhang (2006) for a contemporary review of the use of range based estimators in the context of futures.

market. Daigler (1997) observes a U-shaped intraday volatility function for gold futures. In order to identify intraday periodicity, we convert realized variance to realized standard deviation by taking a square root of the periodic variance. Thus volatility for any intraday quarterly interval is computed as  $Sqrt\left(\sum_{t=1..21} r_t^2\right)$ .

Graphs 1 and 2 represent the intraday evolution of hourly volatility for gold and crude oil futures respectively. We plot hourly volatility measured as the square root of sum of return squared for each one hour interval over hours of the day. Each point on the graph represents the arithmetic mean of realized standard deviation for the respective preceding one hour interval. The arithmetic mean is computed for the standard deviation for the respective interval over 728 trading days. The hourly volatility plots indicate a ladle shape- a light positive trend during the first half of trading and an inverse 'U' shape over the later part of the day. Volatility begins climbing around 17:00 hours and peaks between 20:00 and 21:00 hours, around 9 p.m. (Indian time) immediately followed by a sharp drop perhaps with a similar drop in trading volume for afterhours trading. Surprisingly there is no overnight spike in volatility but an early evening surge in volatility that does not seem to be in response to US market trading but perhaps another Asian market, most likely Osaka or Hong Kong. This inverse 'U' shape is markedly different than a 'U' shape observed by Daigler (1997) for gold futures in the US. The fact that both gold and oil futures show similar intraday patterns, it is likely that this is systemic and the surge and drop in volatility and presumably volume too might also be due to a lead lag relation with domestic spot markets rather than international markets.

We also conduct formal parametric and non-parametric tests for intraday effects on volatility. For parametric testing, we use a simple t-test for pair wise difference in means assuming unequal variance. We report the t-test results in Tables 3. The two top panels in Table 3 contain the difference in mean and the corresponding significance level in notation for gold futures; the two bottom panels contain similar results for the crude oil futures.

For non-parametric testing, we use the JT test as in Duong and Kaley (2008) to test the null that volatility for all time intervals are similar. The JT test statistic is computed as follows. First, we identify the  $k$ th intra day time interval for the  $i$ th futures where  $k=1$  identifies the first quarter (approximately 21 10 minutes intervals) from opening of trading day and  $k=2$  identifies the second quarter (22-42 10 minutes intervals) after opening and so on. We pair each observation ( $RV1$ ) from  $k=1$  with that from  $k^{-1}$  ( $k=2, 3, \dots, K$ ) and compute the difference. We then compute  $U_{ik}$  as follows: If  $RV1 > RV2$  then  $U_{ik} = 1$ , if  $RV1 < RV2$  then  $U_{ik} = -1$ , if  $RV1 = RV2$  then  $U_{ik} = 0$ .<sup>9</sup> The test statistic

$J = \sum_{k=1}^{K-1} U_{ik}$  converges to  $Z(0,1)$ , standard normal distribution for a large sample size.

Table 3B contains results from the JT test for gold and crude oil futures. The high z-values (83 and 81 with p-values  $< 0.000$ ) reject the null hypothesis that volatility is identical in each quarter of the day.

Further we test for seasonality i.e., day of the week and months of the year effects on our computed volatility measures, sum of squared returns and Parkinson's range

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<sup>9</sup> We redefine the random variable  $U_{ik}$  to assume values -1, +1, and 0 to center it on zero and thus increase the power of the test. In the original formulation where the values of the random variable are 1, 0.5, and 0, the power of the JT test is low since the lower bound is set at 0, the center is at 0.5, and hence the probability of a positive value of the statistic is close to 1.

estimate in a regression framework. Tables 4A and 4B report the parameter estimates and the corresponding t-statistics based on Newey-West adjusted standard errors for gold and oil futures volatility estimates respectively. Model 1 identifies a day-of-the week effect in volatility. The day of the week effect is carried out by introducing a dummy variable denoting each day of the week, while the intercept denoting the missing day of the week, in this case Saturday. The significant day of the week negative coefficients for all weekdays, Monday through Friday for both gold and crude oil volatility imply that Saturday volatility is higher than that of any weekday. These findings are not surprising given the descriptive statistics of weekday and Saturday volatility in Tables 2A-2B.

Model 2 combines month of the year effect with the day of the week effect. We exclude Saturday observations from both gold and crude oil futures volatility and hence the reduction in sample size from 869 to 728. We report the estimated parameters and their corresponding t-statistics based on a consistent standard error in Tables 4A and 4B, columns 3 in each table for gold and oil futures respectively. The results indicate there is no day of the week but a strong month of the year effect, which is different between gold and oil futures.

We identify two different monthly seasonality effects for gold and oil futures- a bimonthly effect for two consecutive months, September and October for gold futures; and a quarter of the year effect for crude oil futures, that becomes apparent in the last quarter of the year. To provide conclusive evidence, we test the hypothesis by creating a bimonthly dummy for each two successive months beginning with January for gold futures and similarly creating four quarter (in a year) dummies for oil futures and run a

regression model with bimonthly and quarter dummies as regressor variables for gold and oil futures respectively. The results are contained in Table 4, column 4 & 7. Later, we include the bimonthly and quarter-of-the-year dummies as controls for a robustness test of Samuelson's maturity hypothesis.

#### IV. Testing Samuelson's Maturity Hypothesis: Methodology and Results

We test Samuelson's hypothesis on the relation between return volatility and time to maturity via several regression models. Samuelson's hypothesis postulates: *Futures price volatility increases as futures approach expiration i.e., time to maturity decreases.*

To test Samuelson's hypothesis, we propose the following basic regression model:

$$Model: Vol_t = \alpha + \beta_1 * TTM + \beta_2 * \sqrt{TTM} + \varepsilon_t \dots (1)$$

where,  $Vol_t$  = Daily realized volatility

$TTM$  = number of days to maturity

We use two measures of daily volatility- one, the average of sum of intraday squared returns for each 10 minutes interval and two, Parkinson's range based measure based on intraday bid and ask quotes. Regarding independent variables in the regression, besides  $TTM$ , we introduce a square root ( $\sqrt{TTM}$ ) term as an additional independent variable to adjust for a possible nonlinear trend in volatility.

The OLS regression results are contained in Table 5. Model 1 reports the parameter estimates along with the t-statistics and the corresponding significance for the

basic regression model. In Model 2, we control for seasonality by including the bi-month effect for the gold and quarter-of-the-year effect for crude oil respectively, as control variables. The results are contained under Model 2.

We further include spot volatility in the above regression model to test the explanatory power of TTM after controlling for quarter-of-the-year and spot market information flow. In order to compute spot volatility, we need spot prices. We obtain intraday one minute interval spot price for gold and oil from NCDEX. To be consistent, we arrange the one minute interval data to ten minutes interval and from those prices compute spot volatility as the sum of squared returns. Model 3 contains the results.

We further expand the model to test its robustness by including quarter-of-the year effects, spot volatility, and also log (Average |buy quantity – sell quantity|) as a proxy for net order flow in a comprehensive model as follows.

$$Vol_t = \alpha + \beta_1 * TTM + \beta_2 * \sqrt{TTM} + \beta_3 * Spotvol + \beta_4 * Netflow + \sum_j \varphi_j \delta_j + \varepsilon_t \dots (2)$$

where, TTM = number of days to maturity

$$\text{Sqrt (TTM)} = (TTM)^{1/2}$$

$\delta_j$  = bi-month or quarter dummy denoting each six bi-month and four quarters in a year for the gold and oil respectively

$$\text{Spotvol} = \text{spot volatility}$$

$$\text{Netflow} = \text{Log (Average (Abs (buy-sell))}$$

Table 5, Model 4 contains the results of our comprehensive model. The significant coefficient estimates indicate that a net inverse non-linear relation exists between realized volatility and time to expiration is and that this relation is robust.

However, in terms of explanatory power, spot volatility combined with net order flow denoting information and inventory effect clearly dominate TTM as shown by the adjusted R-squares in models 2 and 4. The  $R^2$  for oil futures increases from 5% in model 2 to 40% in model 4 while the same for gold futures increases from 7% in model 2 to 60% in model 4.

The results in Table 5 show strong support for the extended model for testing Samuelson's maturity hypothesis. Nevertheless, we would like to know whether such support for the maturity hypothesis is due to the carrying cost associated with non-financial futures. Hence we test the negative covariance hypothesis proposed by Bessembinder (1986) for both gold and crude oil futures by the following regression model.

$$\text{Model: } \Delta C_t = \alpha_0 + \alpha_1 * \Delta S_t + \varepsilon_t$$

$$\text{where } C_t = \frac{F_t - S_t}{TTM_t}$$

$$F_t = \log(\text{futures price at time } t)$$

$$S_t = \log(\text{Spot price at time } t)$$

In order to test the negative covariance hypothesis, we need spot prices. As mentioned earlier, we obtain intraday one minute interval spot price for gold and oil futures from NCDEX and arrange the one minute interval data to ten minutes interval and from those prices compute spot volatility as the sum of squared returns.

Table 6 reports the parameter estimates and the corresponding t-statistics for the regression model above testing the negative covariance hypothesis. Obviously we expect a positive  $\alpha$  and a strictly negative  $\beta$  and thus a one-tail t-test is appropriate in this

instance. We find  $\beta$  (gold -831, crude oil -593) to be significantly (at less than 1%, and 10%) negative for both gold and crude oil futures. The support for negative covariance hypothesis, albeit one strongly and the other weakly with respect to gold and oil futures volatility as evidenced by the results in Table 6 indicates that carrying cost is perhaps an essential component of those two futures. However, contrary to a linear relation, carrying cost seems to have an asymmetric effect on volatility. More importantly though the results indicate a complex dynamics between carrying cost and uncertainty arising from inventory and information exposure, which at the end determine gold and oil futures volatility.

## **V. Conclusion**

We compute and examine two different estimates of realized volatility, sum of squared returns and Parkinson's range based estimate, from intraday data on crude oil and gold futures trading on MCX, the leading commodities exchange in India. First, we test for and identify any day-of-the-week and month-of-the-year effects in realized volatility. Second, we test Samuelson's hypothesis that volatility increases as the contract draws nearer to expiration after controlling for seasonality, spot volatility, and net order flow denoting information and inventory uncertainty. Our empirical results strongly support a net non-linear inverse relation between realized volatility and time to expiration. However, information and inventory uncertainty explain the better part of the variation in realized volatility estimates of both gold and crude oil futures.

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Table 1: The current state of trading of most liquid commodity futures in India

| Category/<br>commodity | Trading<br>venue | Unit | Year<br>started | Contract<br>expiration<br>month | Daily<br>average<br>volume | % Share<br>of total of<br>the<br>exchange<br>(by<br>value*) |
|------------------------|------------------|------|-----------------|---------------------------------|----------------------------|-------------------------------------------------------------|
| <b>Precious Metals</b> |                  |      |                 |                                 |                            |                                                             |
| Gold                   | MCX              | KG   | 2003            | F,A,J,A,O,D                     | 44751                      | 34.88                                                       |
| Silver                 | MCX              | MT   | 2003            | J,M,M,M,S,N                     | 1415                       | 17.31                                                       |
| Copper                 | MCX              | MT   | 2004            | F,A,J,A,O,D                     | 97057                      | 12.55                                                       |
| Zinc                   | MCX              | MT   | 2006            | F,A,J,A,O,D                     | 66120                      | 2.87                                                        |
| Nickel                 | MCX              | MT   | 2004            | F,A,J,A,O,D                     | 8027                       | 3.19                                                        |
| <b>Energy</b>          |                  |      |                 |                                 |                            |                                                             |
| Crude Oil              | MCX              | BBL  | 2005            | J,F,M,...D                      | 13473390                   | 20.32                                                       |
| <b>Agricultural</b>    |                  |      |                 |                                 |                            |                                                             |
| Cumin (Jeera)          | NCDEX            | MT   | 2005            | J,F,M,...D                      | 7193                       | 3.45                                                        |
| Guar seed              | NCDEX            | MT   | 2004            | J,F,M,...D                      | 297327                     | 24.77                                                       |
| Pepper                 | NCDEX            | MT   | 2004            | J,F,M,...D                      | 1484                       | 2.73                                                        |
| Soy oil                | NCDEX            | MT   | 2004            | J,F,M,...D                      | 83708                      | 14.92                                                       |
| Soy oil                | MCX              | MT   | 2004            | J,F,M,...D                      | 12469                      | 0.29                                                        |
| Chick peas (Chana)     | NCDEX            | MT   | 2004            | J,F,M,...D                      | 118182                     | 10.92                                                       |
| R/M seed               | NCDEX            | MT   | 2003            | J,F,M,...D                      | 77860                      | 9.49                                                        |

\* These figures represents percentage share of the specific commodity in value in the specific exchange during the year 2009.

The total value of traded contracts in the year 2009 for MCX is INR 59.57 trillion and in NCDEX is INR 8.04 trillion.

For the year 2009, the top two commodities in terms of value of traded contracts in MCX are gold (34.88%) and crude oil (20.88%).

Table 2A: Descriptive statistics of the gold futures price volatility

|                 | volatility measure     |           |                         |           |
|-----------------|------------------------|-----------|-------------------------|-----------|
|                 | Sum of squared returns |           | Parkinson's range based |           |
|                 | Week days              | Saturdays | Week days               | Saturdays |
| Mean            | -3.9905                | -7.7067   | -4.36                   | -8.3606   |
| 25th Percentile | -4.6957                | -8.7535   | -5.1827                 | -9.569    |
| 50th Percentile | -3.9179                | -7.8044   | -4.2858                 | -8.4132   |
| 75th Percentile | -3.3047                | -7.0141   | -3.5074                 | -7.2994   |
| Std Dev         | 1.0859                 | 1.3269    | 1.3097                  | 1.5938    |
| Skewness        | -0.4027                | 0.8934    | -0.3383                 | 0.2564    |
| Kurtosis        | 4.1756                 | 4.7954    | 4.0178                  | 2.6587    |
| N               | 728                    | 141       | 728                     | 141       |

Descriptive statistics of the natural log of daily realized volatility (annualized), estimated for the gold futures contracts closest to maturity. Intraday returns have been calculated by using mid quotes. Realized volatility is estimated by using the sum of squared returns and Parkinson's range based measure. Two samples Kolmogorov Smirnov test is applied to test the difference between the distributions of realized volatility for the week days and Saturdays. KS test provides D value--0.8739 with p value less than 0.000 for the realized volatility estimated as the sum of squared returns and D-value 0.8405 with p-value less than 0.000 for the volatility estimated as the Parkinson's measure, rejects the null hypothesis, that the distribution of the log of realized volatility is same for the Saturdays and for the week days in both cases.

Table 2B: Descriptive statistics of the crude oil futures price volatility

|                 | volatility measure     |           |                         |           |
|-----------------|------------------------|-----------|-------------------------|-----------|
|                 | Sum of squared returns |           | Parkinson's range based |           |
|                 | Week days              | Saturdays | Week days               | Saturdays |
| Mean            | -2.997                 | -7.169    | -3.3752                 | -7.9032   |
| 25th Percentile | -3.455                 | -7.871    | -4.0058                 | -8.6206   |
| 50th Percentile | -3.048                 | -7.293    | -3.3555                 | -7.9201   |
| 75th Percentile | -2.497                 | -6.511    | -2.6784                 | -7.06     |
| Std Dev         | 0.983                  | 1.1       | 1.1811                  | 1.2951    |
| Skewness        | -1.056                 | 0.211     | -0.8312                 | -0.0368   |
| Kurtosis        | 8.009                  | 2.89      | 6.6181                  | 3.001     |
| N               | 728                    | 133       | 728                     | 133       |

Descriptive statistics of the natural log of daily realized volatility (annualized), estimated for the crude oil futures contracts closest to maturity. Intraday returns have been calculated by using mid quotes. Realized volatility is estimated by using the sum of squared returns and Parkinson's range based measure. Two samples Kolmogorov Smirnov test is applied to test the difference between the distributions of realized volatility for the week days and Saturdays. KS test provides D value--0.9560 (p-value < 0.000) for the realized volatility estimated as the sum of squared returns and D-value 0.9519 (p-value < 0.000) for the volatility estimated as the Parkinson's measure, rejects the null hypothesis, that the distribution of the log of realized volatility is same for the Saturdays and for the week days in both cases.

Table 3A1: Intraday periodicity in realized volatility (difference in means) for crude oil futures

| To<br>From | 12:10   | 13:10   | 14:10   | 15:10   | 16:10   | 17:10   | 18:10   | 19:10   | 20:10   | 21:10   | 22:10   | 23:10   | 23:55   |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 11:10      | -0.001* | -0.065* | -0.123* | -0.130* | -0.135* | -0.135* | -0.167* | -0.276* | -0.385* | -0.423* | -0.367* | -0.341* | -0.164* |
| 12:10      |         | -0.064* | -0.121* | -0.128* | -0.134* | -0.134* | -0.166* | -0.275* | -0.384* | -0.421* | -0.366* | -0.34*  | -0.162* |
| 13:10      |         |         | -0.058* | -0.064* | -0.07*  | -0.07*  | -0.102* | -0.211* | -0.32*  | -0.357* | -0.302* | -0.276* | -0.098* |
| 14:10      |         |         |         | -0.007  | -0.012  | -0.012  | -0.044* | -0.153* | -0.262* | -0.300* | -0.245* | -0.218* | -0.041* |
| 15:10      |         |         |         |         | -0.005  | -0.005  | -0.037* | -0.146* | -0.255* | -0.293* | -0.238* | -0.211* | -0.034* |
| 16:10      |         |         |         |         |         | 0.000   | -0.032* | -0.141* | -0.250* | -0.288* | -0.232* | -0.206* | -0.029^ |
| 17:10      |         |         |         |         |         |         | -0.032* | -0.141* | -0.250* | -0.288* | -0.232* | -0.206* | -0.029^ |
| 18:10      |         |         |         |         |         |         |         | -0.109* | -0.218* | -0.256* | -0.200* | -0.174* | 0.003   |
| 19:10      |         |         |         |         |         |         |         |         | -0.109* | -0.147* | -0.091* | -0.065* | 0.112*  |
| 20:10      |         |         |         |         |         |         |         |         |         | -0.038^ | 0.018   | 0.044^  | 0.221*  |
| 21:10      |         |         |         |         |         |         |         |         |         |         | 0.055*  | 0.081*  | 0.259*  |
| 22:10      |         |         |         |         |         |         |         |         |         |         |         | 0.026   | 0.204*  |
| 23:10      |         |         |         |         |         |         |         |         |         |         |         |         | 0.177*  |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. Table show the mean difference between the hourly realized volatility (multiplied by 100) estimated as the square root of sum of squared returns of ten minute intervals for 728 days. From 14:10 hrs to 23:10 hrs means that the difference between the hourly realized volatility estimated from 13:10 hrs to 14:10 hrs & 22:10 hrs to 23:10 hrs. Paired two sample t-test is used to test the significance of the difference in means.

Table 3A2: Intraday periodicity in realized volatility (difference in means) for crude oil futures

| To<br>From | 12:10 | 13:10   | 14:10   | 15:10   | 16:10   | 17:10   | 18:10   | 19:10   | 20:10   | 21:10   | 22:10   | 23:10   | 23:55   |
|------------|-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|            | -     |         |         |         |         |         |         |         |         |         |         |         |         |
| 11:10      | 0.006 | -0.056* | -0.095* | -0.095* | -0.095* | -0.092* | -0.114* | -0.195* | -0.266* | -0.28*  | -0.241* | -0.227* | -0.066* |
| 12:10      |       | -0.049* | -0.088* | -0.089* | -0.089* | -0.085* | -0.107* | -0.189* | -0.259* | -0.274* | -0.234* | -0.221* | -0.06*  |
| 13:10      |       |         | -0.039* | -0.04*  | -0.04*  | -0.036* | -0.058* | -0.14*  | -0.21*  | -0.225* | -0.185* | -0.171* | -0.011  |
| 14:10      |       |         |         | -0.001  | -0.001  | 0.003   | -0.019^ | -0.101* | -0.171* | -0.186* | -0.146* | -0.132* | 0.028*  |
| 15:10      |       |         |         |         | 0       | 0.004   | -0.018^ | -0.1*   | -0.17*  | -0.185* | -0.146* | -0.132* | 0.029*  |
| 16:10      |       |         |         |         |         | 0.003   | -0.018^ | -0.1*   | -0.17*  | -0.185* | -0.146* | -0.132* | 0.029*  |
| 17:10      |       |         |         |         |         |         | -0.022^ | -0.104* | -0.174* | -0.189* | -0.149* | -0.135* | 0.026*  |
| 18:10      |       |         |         |         |         |         |         | -0.082* | -0.152* | -0.167* | -0.127* | -0.113* | 0.047*  |
| 19:10      |       |         |         |         |         |         |         |         | -0.07*  | -0.085* | -0.045* | -0.031^ | 0.129*  |
| 20:10      |       |         |         |         |         |         |         |         |         | -0.015  | 0.025#  | 0.039*  | 0.199*  |
| 21:10      |       |         |         |         |         |         |         |         |         |         | 0.039*  | 0.053*  | 0.214*  |
| 22:10      |       |         |         |         |         |         |         |         |         |         |         | 0.014   | 0.175*  |
| 23:10      |       |         |         |         |         |         |         |         |         |         |         |         | 0.161*  |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. Table show the mean difference between the hourly realized volatility (multiplied by 100) estimated as the square root of Parkinson's range based measure for 728 days. From 14:10 hrs to 23:10 hrs means that the difference between the hourly realized volatility estimated from 13:10 hrs to 14:10 hrs & 22:10 hrs to 23:10 hrs. Paired two sample t-test is used to test the significance of the difference in means.

Table 3A3: Intraday periodicity in realized volatility (difference in means) for gold futures

| To<br>From | 12:10 | 13:10   | 14:10   | 15:10   | 16:10   | 17:10   | 18:10   | 19:10   | 20:10   | 21:10   | 22:10   | 23:10   | 23:55  |
|------------|-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|--------|
| 11:10      | 0.008 | -0.014# | -0.022* | -0.009  | -0.013# | -0.024* | -0.094* | -0.155* | -0.189* | -0.158* | -0.126* | -0.099* | 0.028* |
| 12:10      |       | -0.022* | -0.03*  | -0.017^ | -0.021* | -0.033* | -0.102* | -0.164* | -0.198* | -0.166* | -0.134* | -0.107* | 0.019* |
| 13:10      |       |         | -0.008  | 0.005   | 0.001   | -0.011  | -0.08*  | -0.142* | -0.176* | -0.144* | -0.112* | -0.085* | 0.042* |
| 14:10      |       |         |         | 0.013#  | 0.009   | -0.003  | -0.072* | -0.134* | -0.168* | -0.136* | -0.104* | -0.077* | 0.049* |
| 15:10      |       |         |         |         | -0.004  | -0.016^ | -0.085* | -0.147* | -0.181* | -0.149* | -0.117* | -0.09*  | 0.037* |
| 16:10      |       |         |         |         |         | -0.012  | -0.081* | -0.143* | -0.177* | -0.145* | -0.113* | -0.086* | 0.041* |
| 17:10      |       |         |         |         |         |         | -0.07*  | -0.131* | -0.165* | -0.134* | -0.102* | -0.074* | 0.052* |
| 18:10      |       |         |         |         |         |         |         | -0.061* | -0.095* | -0.064* | -0.032* | -0.005  | 0.122* |
| 19:10      |       |         |         |         |         |         |         |         | -0.034^ | -0.002  | 0.03^   | 0.057*  | 0.183* |
| 20:10      |       |         |         |         |         |         |         |         |         | 0.031^  | 0.064*  | 0.091*  | 0.217* |
| 21:10      |       |         |         |         |         |         |         |         |         |         | 0.032*  | 0.059*  | 0.186* |
| 22:10      |       |         |         |         |         |         |         |         |         |         |         | 0.027^  | 0.154* |
| 23:10      |       |         |         |         |         |         |         |         |         |         |         |         | 0.127* |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. Table show the mean difference between the hourly realized volatility (multiplied by 100) estimated as the square root of sum of squared returns for 728 days. From 14:10 hrs to 23:10 hrs means that the difference between the hourly realized volatility estimated from 13:10 hrs to 14:10 hrs & 22:10 hrs to 23:10 hrs. Paired two sample t-test is used to test the significance of the difference in means.

Table 3A4: Intraday periodicity in realized volatility (difference in means) for crude oil futures

| To<br>From | 12:10  | 13:10   | 14:10   | 15:10  | 16:10   | 17:10   | 18:10   | 19:10   | 20:10   | 21:10   | 22:10   | 23:10   | 23:55  |
|------------|--------|---------|---------|--------|---------|---------|---------|---------|---------|---------|---------|---------|--------|
| 11:10      | -0.001 | -0.014^ | -0.016* | -0.007 | -0.013^ | -0.014* | -0.071* | -0.102* | -0.12*  | -0.099* | -0.077* | -0.064* | 0.033* |
| 12:10      |        | -0.014^ | -0.016* | -0.007 | -0.013^ | -0.014* | -0.07*  | -0.102* | -0.12*  | -0.099* | -0.077* | -0.063* | 0.033* |
| 13:10      |        |         | -0.002  | 0.007  | 0.001   | 0       | -0.057* | -0.088* | -0.106* | -0.085* | -0.063* | -0.05*  | 0.047* |
| 14:10      |        |         |         | 0.009  | 0.003   | 0.002   | -0.055* | -0.086* | -0.104* | -0.083* | -0.061* | -0.048* | 0.049* |
| 15:10      |        |         |         |        | -0.006  | -0.007  | -0.064* | -0.095* | -0.113* | -0.092* | -0.07*  | -0.057* | 0.04*  |
| 16:10      |        |         |         |        |         | -0.001  | -0.058* | -0.089* | -0.107* | -0.086* | -0.064* | -0.05*  | 0.046* |
| 17:10      |        |         |         |        |         |         | -0.057* | -0.088* | -0.106* | -0.085* | -0.063* | -0.049* | 0.047* |
| 18:10      |        |         |         |        |         |         |         | -0.031* | -0.049* | -0.028* | -0.006  | 0.007   | 0.104* |
| 19:10      |        |         |         |        |         |         |         |         | -0.018# | 0.003   | 0.025*  | 0.039*  | 0.135* |
| 20:10      |        |         |         |        |         |         |         |         |         | 0.021^  | 0.043*  | 0.057*  | 0.153* |
| 21:10      |        |         |         |        |         |         |         |         |         |         | 0.022*  | 0.035*  | 0.132* |
| 22:10      |        |         |         |        |         |         |         |         |         |         |         | 0.014   | 0.11*  |
| 23:10      |        |         |         |        |         |         |         |         |         |         |         |         | 0.097* |

Note: \*, ^, and # denote significance at 1%, 5%, and 10% levels respectively. The number in each cell denotes the mean difference between the hourly realized volatility (multiplied by 100) estimated as the square root of Parkinson's range based measure for 728 days e.g., 14:10 hrs to 23:10 hrs means that the difference between the hourly realized volatility estimated from 13:10 hrs to 14:10 hrs & 22:10 hrs to 23:10 hrs. Paired two sample t-test is used to test the significance of the difference in means.

Table 3B: Intraday periodicity test: JT-test

|             | Crude Oil | Gold    |
|-------------|-----------|---------|
| Z Statistic | -83.8345  | 83.8148 |
| (p-value)   | 0.0000    | 0.0000  |
| RV1.2       | -0.1803   | -0.0264 |
| RV1.3       | -0.4656   | -0.2507 |
| RV1.4       | -0.5454   | -0.1757 |
| RV2.3       | -0.2715   | -0.2144 |
| RV2.4       | -0.3615   | -0.1320 |
| RV3.4       | -0.0617   | 0.0709  |

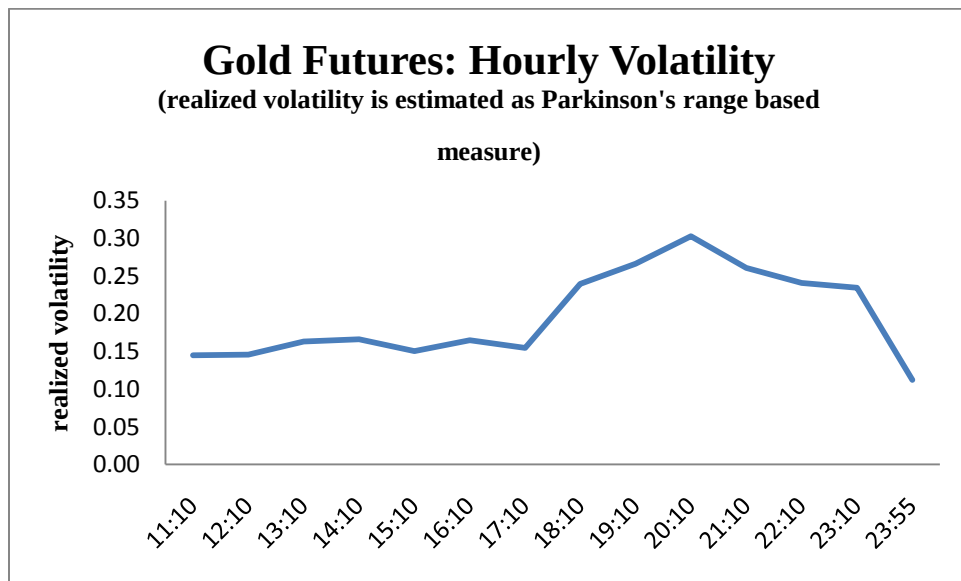
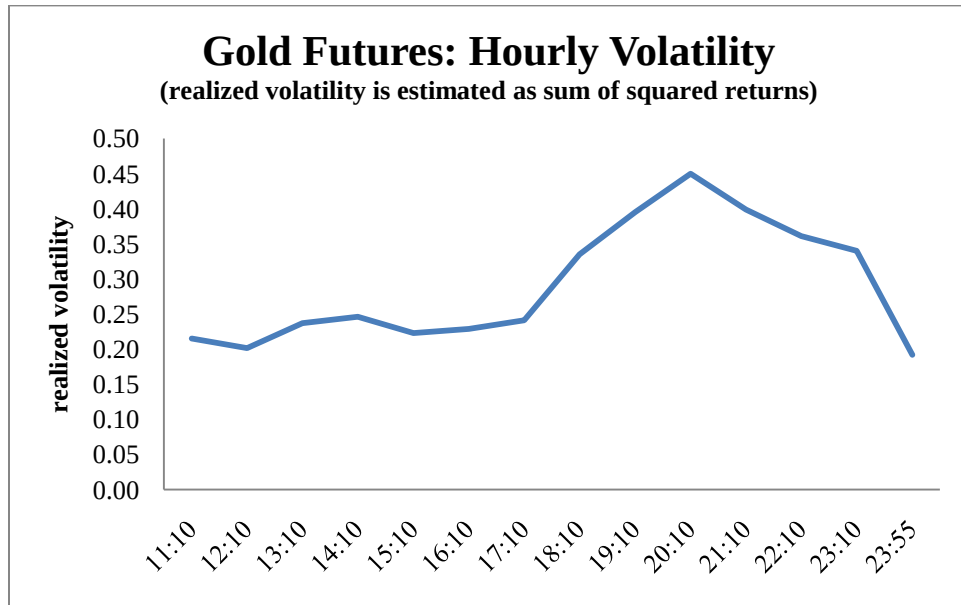
Calculation is based on 100 times the square root of the realized volatility over each of the 4 quarters of a trading Day. Quarter 1 = 10:20 Hrs – 13:30 Hrs, Quarter 2 = 13:40 Hrs – 17:00 Hrs, Quarter 3 = 17:10 Hrs – 20:30 Hrs, Quarter 4 = 20:40 Hrs – 23:55 Hrs.

Number of Trading Days Covered: 728

$RV_{i,j}$  = Difference between the volatility in the  $i$ th and  $j$ th Quarter of the day,  $i = 1,2,3$  and  $j = 2,3,4$

Each Cell represents the median of  $RV_{i,j}$  for each pair  $(i, j)$

Graph 1: Intraday volatility for gold futures



Graph 2: Crude Oil futures volatility

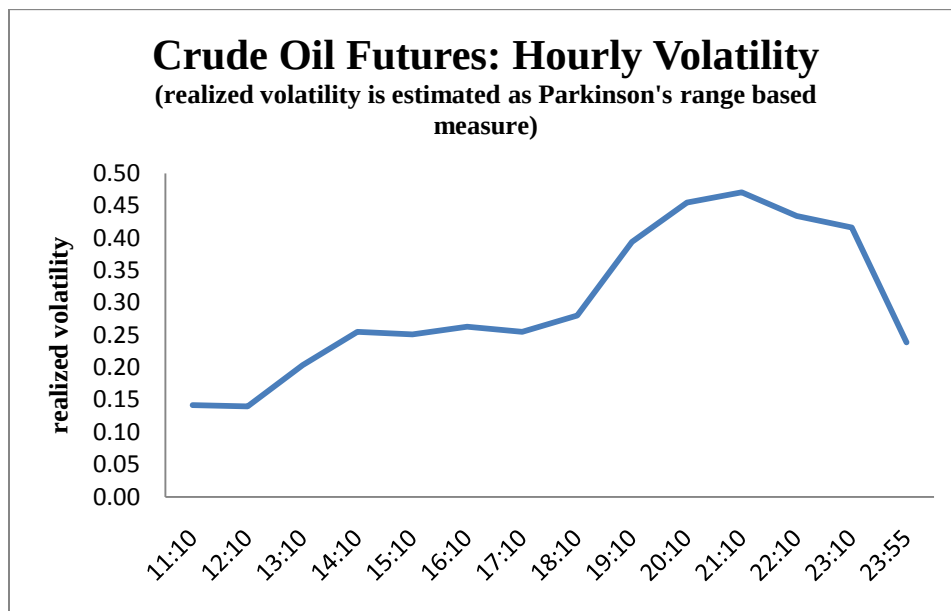
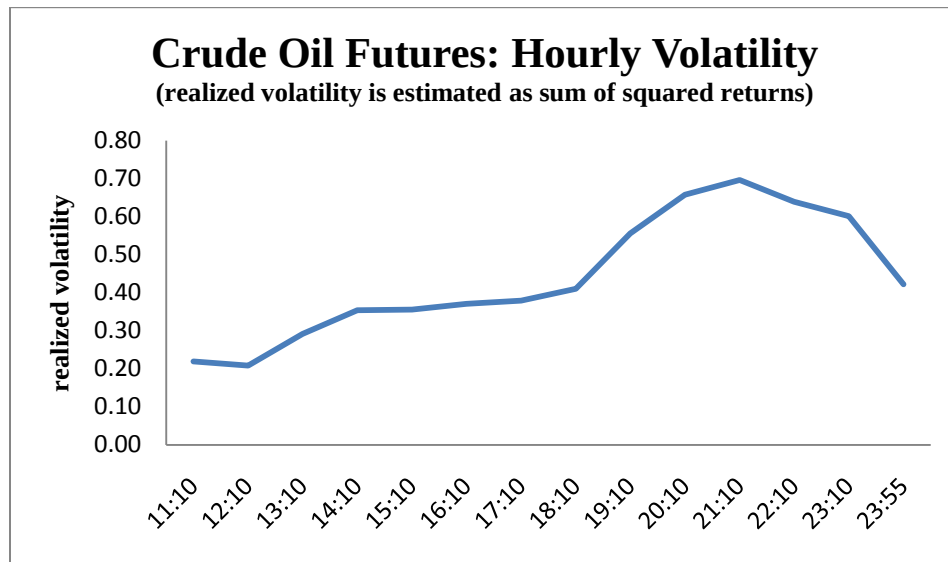


Table 4A: Day of the week and month of the year effects in gold futures volatility

| (1)          | Squared Returns |                 |                | Parkinson's measure |                |                |
|--------------|-----------------|-----------------|----------------|---------------------|----------------|----------------|
|              | (2)             | (3)             | (4)            | (5)                 | (6)            | (7)            |
| Intercept    | 0.0009(0.67)    | 0.0113*(4.15)   | 0.0111*(7.02)  | 0.0003(0.20)        | 0.0073^(1.99)  | 0.0087*(4.10)  |
| Monday       | 0.0103*(5.60)   | -0.0030(-1.58)  |                | 0.0092*(3.82)       | -0.0030(-1.16) |                |
| Tuesday      | 0.0099*(5.40)   | -0.0034#(-1.77) |                | 0.0102*(4.22)       | -0.0021(-0.80) |                |
| Wednesday    | 0.0129*(7.03)   | -0.0005(-0.26)  |                | 0.0125*(5.19)       | 0.0002(0.09)   |                |
| Thursday     | 0.0130*(7.07)   | -0.0003(-0.13)  |                | 0.0116*(4.81)       | -0.0005(-0.20) |                |
| Friday       | 0.0133*(7.24)   |                 |                | 0.0122*(5.04)       |                |                |
| January      |                 | -0.0007(-0.20)  |                |                     | 0.0022(0.48)   |                |
| February     |                 | -0.0030(-0.91)  |                |                     | -0.0004(-0.08) |                |
| March        |                 | -0.0013(-0.41)  |                |                     | 0.0023(0.55)   |                |
| April        |                 | 0.0040(1.23)    |                |                     | 0.0080#(1.86)  |                |
| May          |                 | 0.0053#(1.69)   |                |                     | 0.0070#(1.65)  |                |
| June         |                 | 0.0040(1.26)    |                |                     | 0.0061(1.43)   |                |
| July         |                 | 0.0025(0.78)    |                |                     | 0.0042(0.98)   |                |
| August       |                 | 0.0006(0.19)    |                |                     | 0.0027(0.64)   |                |
| September    |                 | 0.0064^(2.01)   |                |                     | 0.0095^(2.20)  |                |
| October      |                 | 0.0121*(3.79)   |                |                     | 0.0142*(3.31)  |                |
| November     |                 | 0.0021(0.64)    |                |                     | 0.0042(0.98)   |                |
| Jan-Feb      |                 |                 | -0.0033(-1.45) |                     |                | -0.0018(-0.58) |
| Mar-Apr      |                 |                 | -0.0000(-0.01) |                     |                | 0.0026(0.91)   |
| May-Jun      |                 |                 | 0.0035(1.62)   |                     |                | 0.0041(1.43)   |
| Jul-Aug      |                 |                 | 0.0003(0.13)   |                     |                | 0.0010(0.34)   |
| Sep-Oct      |                 |                 | 0.0080*(3.73)  |                     |                | 0.0094*(3.25)  |
| No. of Obs.  | 869             | 728             | 728            | 869                 | 728            | 728            |
| F-stat       | 15.088          | 3.182           | 6.66           | 7.569               | 1.744          | 3.778          |
| p-value      | 0               | 0               | 0              | 0                   | 0.039          | 0.002          |
| Adj.R-square | 0.075           | 0.043           | 0.037          | 0.036               | 0.015          | 0.019          |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. %. Regression Coefficients of the dummies for the day, month and bimonth are given in the table to capture the week of the day and month of the year effect with the realized volatility as the dependent variable, value of the t-statistic is given the parentheses.

Table 4B: Day of week, month of the year effects for crude oil futures volatility

| (1)           | Squared Returns |                 |                 | Parkinson's measure |                 |                 |
|---------------|-----------------|-----------------|-----------------|---------------------|-----------------|-----------------|
|               | (2)             | (3)             | (4)             | (5)                 | (6)             | (7)             |
| Intercept     | 0.0006(0.21)    | 0.0491*(9.35)   | 0.0510*(20.38)  | 0.0003(0.11)        | 0.0437*(7.36)   | 0.0432*(15.24)  |
| Monday        | 0.0285*(7.39)   | -0.0002(-0.04)  |                 | 0.0285*(6.70)       | 0.0072(1.63)    |                 |
| Tuesday       | 0.0286*(7.38)   | 0.0004(0.09)    |                 | 0.0257*(6.01)       | 0.0048(1.08)    |                 |
| Wednesday     | 0.0339*(8.75)   | 0.0049(1.26)    |                 | 0.0248*(5.80)       | 0.0032(0.73)    |                 |
| Thursday      | 0.0297*(7.65)   | 0.0014(0.36)    |                 | 0.0247*(5.75)       | 0.0036(0.80)    |                 |
| Friday        | 0.0290*(7.46)   |                 |                 | 0.0217*(5.05)       |                 |                 |
| January       |                 | -0.0252*(-3.91) |                 |                     | -0.0272*(-3.72) |                 |
| February      |                 | -0.0321*(-5.07) |                 |                     | -0.0323*(-4.49) |                 |
| March         |                 | -0.0286*(-4.57) |                 |                     | -0.0286*(-4.04) |                 |
| April         |                 | -0.0301*(-4.76) |                 |                     | -0.0330*(-4.61) |                 |
| May           |                 | -0.0297*(-4.78) |                 |                     | -0.0294*(-4.18) |                 |
| June          |                 | -0.0259*(-4.14) |                 |                     | -0.0266*(-3.75) |                 |
| July          |                 | -0.0260*(-4.19) |                 |                     | -0.0273*(-3.88) |                 |
| August        |                 | -0.0240*(-3.83) |                 |                     | -0.0262*(-3.69) |                 |
| September     |                 | -0.0151^(-2.39) |                 |                     | -0.0187*(-2.62) |                 |
| October       |                 | -0.0010(-0.16)  |                 |                     | -0.0095(-1.34)  |                 |
| November      |                 | 0.0027(0.43)    |                 |                     | -0.0024(-0.34)  |                 |
| Quarter 1     |                 |                 | -0.0293*(-8.32) |                     |                 | -0.0250*(-6.26) |
| Quarter 2     |                 |                 | -0.0292*(-8.39) |                     |                 | -0.0253*(-6.42) |
| Quarter 3     |                 |                 | -0.0225*(-6.46) |                     |                 | -0.0198*(-5.04) |
| No. of Obs.   | 861             | 728             | 728             | 861                 | 728             | 728             |
| F-stat        | 19.882          | 6.754           | 31.105          | 11.685              | 4.133           | 18.018          |
| p-value       | 0.000           | 0.000           | 0.000           | 0.000               | 0.000           | 0.000           |
| Adj. R-square | 0.099           | 0.106           | 0.111           | 0.058               | 0.061           | 0.066           |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. %. Regression Coefficients of the dummies for the day, month and quarters are given in the table to capture the week of the day and month of the year effect with the realized volatility as the dependent variable, value of the t-statistic is given the parentheses.

Table 5A : Testing extended maturity hypothesis with Gold futures- volatility is measured as sum of intraday squared returns

|               | Model-1         | Model-2         | Model-3         | Model-4         |
|---------------|-----------------|-----------------|-----------------|-----------------|
| Intercept     | -8.661*(-32.97) | -8.733*(-32.14) | -5.346*(-16.90) | -2.565*(-6.62)  |
| TTM           | 0.026^(2.18)    | 0.027^(2.30)    | 0.034*(3.36)    | -0.023^(-2.16)  |
| sqrt(TTM)     | -0.315*(-2.70)  | -0.323*(-2.82)  | -0.387*(-3.92)  | 0.275^(2.51)    |
| Spotvol       |                 |                 | 0.317*(15.90)   | 0.264*(13.79)   |
| Netflow       |                 |                 |                 | -0.872*(-10.98) |
| Jan-Feb       |                 | -0.213(-1.46)   | -0.141(-1.12)   | -0.030(-0.25)   |
| Mar-Apr       |                 | -0.149(-1.08)   | -0.183(-1.53)   | -0.034(-0.31)   |
| May-Jun       |                 | 0.279^(2.03)    | 0.102(0.86)     | 0.040(0.36)     |
| Jul-Aug       |                 | 0.069(0.50)     | -0.002(-0.02)   | 0.023(0.21)     |
| Sep-Oct       |                 | 0.427*(3.09)    | 0.199#(1.66)    | 0.242^(2.18)    |
| N             | 728             | 728             | 728             | 728             |
| F-stat        | 6.765           | 6.727           | 39.529          | 54.382          |
| p-value       | 0.001           | 0.000           | 0.000           | 0.000           |
| Adj. R-square | 0.016           | 0.052           | 0.298           | 0.398           |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. Value of t-statistic is given in the parentheses.

Table 5B : Testing extended maturity hypothesis with Gold futures- volatility is measured as Parkinson's range measure computed from intraday mid quotes

|               | Model-1         | Model-2         | Model-3        | Model-4        |
|---------------|-----------------|-----------------|----------------|----------------|
| Intercept     | -7.976*(-32.30) | -7.608*(-31.15) | -1.228*(-4.34) | -2.289*(-5.89) |
| TTM           | 0.033(1.44)     | 0.030(1.33)     | 0.026#(1.70)   | 0.032^(2.08)   |
| sqrt(TTM)     | -0.285#(-1.81)  | -0.253#(-1.67)  | -0.208^(-1.98) | -0.250^(-2.39) |
| Spotvol       |                 |                 | 0.834*(28.08)  | 0.844*(28.59)  |
| Netflow       |                 |                 |                | 0.170*(3.94)   |
| Quarter 1     |                 | -0.741*(-7.36)  | -0.272*(-3.79) | -0.263*(-3.72) |
| Quarter 2     |                 | -0.562*(-5.67)  | -0.044(-0.61)  | 0.016(0.23)    |
| Quarter 3     |                 | -0.395*(-3.98)  | 0.005(0.08)    | 0.065(0.92)    |
| N             | 728             | 728             | 728            | 728            |
| F-stat        | 3.324           | 13.322          | 154.657        | 137.444        |
| p-value       | 0.037           | 0.000           | 0.000          | 0.000          |
| Adj. R-square | 0.006           | 0.078           | 0.559          | 0.568          |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. Value of t-statistic is given in the parentheses.

Table 5C : Testing extended maturity hypothesis with Crude oil futures- volatility is measured as sum of intraday squared returns

|               | Model-1         | Model-2         | Model-3         | Model-4        |
|---------------|-----------------|-----------------|-----------------|----------------|
| Intercept     | -9.298*(-29.17) | -9.399*(-28.38) | -5.800*(-14.44) | -3.487*(-6.78) |
| TTM           | 0.017(1.20)     | 0.018(1.28)     | 0.026^(2.01)    | -0.022(-1.53)  |
| sqrt(TTM)     | -0.213(-1.51)   | -0.221(-1.58)   | -0.290^(-2.31)  | 0.261#(1.79)   |
| Spotvol       |                 |                 | 0.337*(13.31)   | 0.293*(11.52)  |
| Netflow       |                 |                 |                 | -0.725*(-6.87) |
| Jan-Feb       |                 | -0.207(-1.16)   | -0.130(-0.81)   | -0.038(-0.24)  |
| Mar-Apr       |                 | -0.123(-0.73)   | -0.158(-1.05)   | -0.035(-0.23)  |
| May-Jun       |                 | 0.378^(2.26)    | 0.190(1.26)     | 0.138(0.95)    |
| Jul-Aug       |                 | 0.080(0.47)     | 0.004(0.03)     | 0.025(0.17)    |
| Sep-Oct       |                 | 0.459*(2.72)    | 0.217(1.42)     | 0.252#(1.71)   |
| No. of Obs.   | 728             | 728             | 728             | 728            |
| F-stat        | 2.235           | 4.422           | 26.963          | 30.756         |
| p-value       | 0.108           | 0.000           | 0.000           | 0.000          |
| Adj. R-square | 0.003           | 0.032           | 0.222           | 0.269          |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. Value of t-statistic is given in the parentheses.

Table 5D : Testing extended maturity hypothesis with Crude oil futures- volatility is measured as Parkinson's measure based on intraday mid quotes

|               | Model-1         | Model-2         | Model-3        | Model-4        |
|---------------|-----------------|-----------------|----------------|----------------|
| Intercept     | -8.491*(-28.52) | -8.091*(-27.20) | -1.377*(-3.53) | -2.970*(-5.56) |
| TTM           | 0.027(0.97)     | 0.023(0.85)     | 0.019(0.92)    | 0.028(1.33)    |
| sqrt(TTM)     | -0.221(-1.16)   | -0.188(-1.02)   | -0.141(-0.97)  | -0.203(-1.41)  |
| Spotvol       |                 |                 | 0.878*(21.47)  | 0.892*(22.01)  |
| Netflow       |                 |                 |                | 0.255*(4.30)   |
| Quarter 1     |                 | -0.771*(-6.29)  | -0.277*(-2.81) | -0.265*(-2.72) |
| Quarter 2     |                 | -0.596*(-4.94)  | -0.050(-0.52)  | 0.039(0.40)    |
| Quarter 3     |                 | -0.462*(-3.83)  | -0.041(-0.43)  | 0.049(0.50)    |
| No. of Obs.   | 728             | 728             | 728            | 728            |
| F-stat        | 1.160           | 9.209           | 89.389         | 81.123         |
| p-value       | 0.314           | 0.000           | 0.000          | 0.000          |
| Adj. R-square | 0.000           | 0.053           | 0.422          | 0.435          |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%. Value of t-statistic is given in the parentheses.

Table 6: Negative covariance Hypothesis: Test for crude oil and gold futures

|               | Gold             | Crude Oil        |
|---------------|------------------|------------------|
| Intercept     | 0.268(0.12)      | -0.100(-0.01)    |
| Spot Price    | -831.117*(-5.23) | -593.905#(-1.73) |
| N             | 727              | 727              |
| F-stat        | 27.308           | 2.984            |
| P-value       | 0.000            | 0.084            |
| Adj. R-square | 0.035            | 0.003            |

Note: (\*) Significant at 1%, (^) Significant at 5% level, (#) Significant at 10%.

